

Parallel computing

Real world applications for parallel computing is weather forecasting.

How it's used -

- weather models involve processing massive datasets (temperature, humidity, pressure and wind) from satellites and ground station.
- The datasets are broken into grid sections of the earth, each section's forecast is calculated simultaneously using parallel computing.

Why is important -

speed. parallel computing reduces processing time.

Accuracy - more data and finer grids improve prediction accuracy, which is only possible through high-performance parallel processing.

Networked systems - online banking

How it's used -

online banking platforms connect multiple systems user interfaces (web/mobile), authentication servers, transactions processing systems and databases. These components communicate over secure networked systems, often spread across data centers.

Why it's important -

- Real-time access - customers can transfer funds, view balance and pay bills instantly from anywhere.
- security and redundancy - Networked system allows for failover and backup, protecting data and ensuring uptime.
- scalability - millions of users access services, the network can balance loads and maintain performance.